

Ex.no:9	MINI PROJECT – OTT PLATFORM CONTENT ANALYTICS DATA VISUALIZATION TECHNIQUES

Domain and Problem Statement:

Domain: Entertainment & OTT Streaming Media Analytics

Problem Statement: A global over-the-top (OTT) streaming platform hosting thousands of Movies and TV Shows across international markets lacks an integrated business intelligence system to evaluate content catalog growth, format distribution (Movies vs. TV Shows), maturity ratings, regional production hubs, and multi-dimensional genre dynamics. Raw content metadata contains missing values, unstructured text lists, and unparsed duration metrics, hindering catalog managers and content acquisition strategists from making data-driven programming decisions. This project transforms raw content metadata into structured business intelligence across Python, SQL Server, Power BI, and Tableau to evaluate catalog trends, geographic distribution, audience demographic ratings, and dynamic cross-dimensional breakdowns.

Dataset Description:

Dataset Name: OTT Platform Content Dataset (Netflix Titles Catalog).

Granularity: Each row in the dataset represents an individual Movie or TV Show title available in the platform's streaming library.

Timeframe Covered: Content release years spanning 1925 to 2021; platform catalog additions spanning 2008 to 2021.

Key Entities & Attributes:

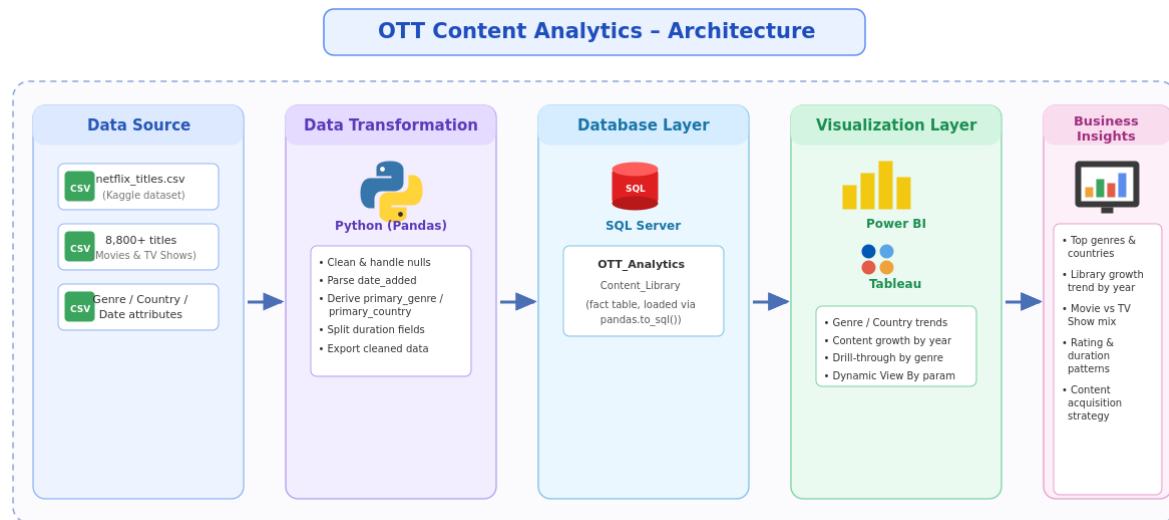
- **Content Catalog (Raw):** show_id, type, title, director, cast, country, date_added, release_year, rating, duration, listed_in, description (8,807 titles).
- **Engineered Attributes:** year_added, month_added, primary_country, primary_genre, duration_value, duration_unit (8,790 cleaned titles).
- **Analytical Dimensions:** Fact_Content (8,790 titles), Dim_Genre (primary genres), Dim_Country (production countries), Dim_Date (temporal addition trends).

Data Source:

Primary Source: Comprehensive streaming catalog metadata extract (netflix_titles.csv – 8,807 titles).

Storage Infrastructure: Cleaned transactional and content catalog hosted within a local SQL Server database instance (OTT_Analytics) under table Content_Library.

Architecture:



Program:

1. Python – Data Cleaning & Transformation

```
import pandas as pd
```

```
from sqlalchemy import create_engine
```

```
# Load dataset
```

```
df = pd.read_csv("netflix_titles.csv")
```

```
ott = df.copy()
```

```
# Handle missing values
```

```
ott["director"] = ott["director"].fillna("Unknown")
```

```
ott["cast"] = ott["cast"].fillna("Unknown")
```

```
ott["country"] = ott["country"].fillna("Unknown")
```

```
ott.dropna(subset=["date_added", "rating", "duration"], inplace=True)
```

```
# Fix date column and engineer temporal features
```

```
ott["date_added"] = pd.to_datetime(ott["date_added"].str.strip())
```

```
ott["year_added"] = ott["date_added"].dt.year
```

```
ott["month_added"] = ott["date_added"].dt.month_name()

# Split multi-value columns — take primary value
ott["primary_country"] = ott["country"].apply(lambda x: x.split(",")[0].strip())
ott["primary_genre"] = ott["listed_in"].apply(lambda x: x.split(",")[0].strip())

# Split duration into numeric fields (Movies = minutes, TV Shows = seasons)
ott["duration_value"] = ott["duration"].str.extract(r"(\d+)").astype(float)
ott["duration_unit"] = ott["duration"].apply(lambda x: "Season(s)" if "Season" in x else "min")

# Remove duplicates and validate integrity
ott.drop_duplicates(subset=["show_id"], inplace=True)

# Load into local SQL Server instance
server = "localhost"
database = "OTT_Analytics"

conn_str =
f'mssql+pyodbc://@{server}/{database}?driver=ODBC+Driver+17+for+SQL+Server&trusted_conn
ection=yes"

engine = create_engine(conn_str)
ott.to_sql("Content_Library", con=engine, if_exists="replace", index=False, chunksize=1000)

# Export cleaned dataset
ott.to_csv("ott_cleaned.csv", index=False)

print("Original Shape:", df.shape)
print("Cleaned Shape:", ott.shape)
ott.head()
```

2. SQL Server – Database and Data Loading

SSMS – New Query window:

```
CREATE DATABASE OTT_Analytics;
```

```
GO
```

```
USE OTT_Analytics;
```

```
GO
```

```
CREATE TABLE Content_Library (  
    [show_id]    VARCHAR(20) PRIMARY KEY,  
    [type]       VARCHAR(20) NOT NULL,  
    [title]      VARCHAR(255) NOT NULL,  
    [director]   VARCHAR(255) NULL,  
    [cast]       VARCHAR(MAX) NULL,  
    [country]    VARCHAR(255) NULL,  
    [date_added] DATE      NOT NULL,  
    [release_year] INT      NOT NULL,  
    [rating]     VARCHAR(20) NOT NULL,  
    [duration]   VARCHAR(50) NOT NULL,  
    [listed_in]  VARCHAR(255) NOT NULL,  
    [description] VARCHAR(MAX) NULL,  
    [year_added] INT      NOT NULL,  
    [month_added] VARCHAR(20) NOT NULL,  
    [primary_country] VARCHAR(100) NULL,  
    [primary_genre] VARCHAR(100) NOT NULL,  
);
```

Verify:

```
SELECT COUNT(*) FROM Content_Library;          -- 8790
```

```
SELECT COUNT(*) FROM Content_Library WHERE type = 'Movie'; -- 6126
```

```
SELECT COUNT(*) FROM Content_Library WHERE type = 'TV Show'; -- 2664
```

Procedure:

Step 1: Data Preparation & Preprocessing (Python / Pandas)

- Loaded the raw netflix_titles.csv dataset into pandas (8,807 rows × 12 columns) and inspected missing values, data distributions, and data types.
- Handled missing attributes by imputing "Unknown" for director (2,634 missing), cast (825 missing), and country (831 missing), and pruned records lacking date_added, rating, or duration.
- Parsed date_added strings into standardized datetime timestamps and engineered temporal attributes (year_added, month_added):

$$\text{total_stays} = \text{stays_in_weekend_nights} + \text{stays_in_week_nights}$$

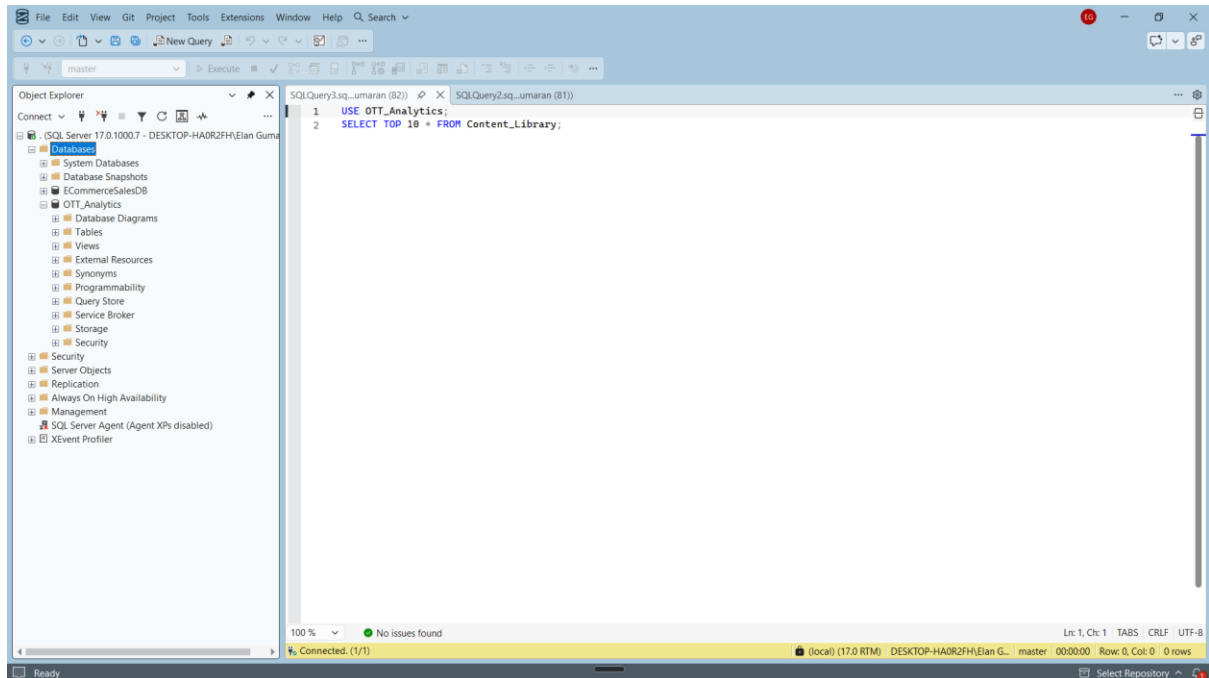
- Engineered calculated columns (primary_country, primary_genre, duration_value, duration_unit), deduplicated on show_id (8,790 records remaining), exported ott_cleaned.csv, and staged data for SQL Server import.

	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_in	description	year_added	month_added	p
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	Unknown	United States	2021-09-25	2020	PG-13	90 min	Documentaries	As her father nears the end of his life, filmm...	2021	September	
1	s2	TV Show	Blood & Water	Unknown	Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban...	South Africa	2021-09-24	2021	TV-MA	2 Seasons	International TV Shows, TV Dramas, TV Mysteries	After crossing paths at a party, a Cape Town t...	2021	September	
2	s3	TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...	Unknown	2021-09-24	2021	TV-MA	1 Season	Crime TV Shows, International TV Shows, TV Act...	To protect his family from a powerful drug lor...	2021	September	
3	s4	TV Show	Jailbirds New Orleans	Unknown	Unknown	Unknown	2021-09-24	2021	TV-MA	1 Season	Docuseries, Reality TV	Feuds, flirtations and toilet talk go down amo...	2021	September	

Step 2: SQL Database Connection & Data Ingestion

- Open Power BI Desktop → Get Data → SQL Server database.

2. Enter the Server (localhost) and Database Name (OTT_Analytics).
3. Select Windows Authentication, choose Import mode, then select Content_Library (Fact_Content) along with dimensional queries (Dim_Genre, Dim_Country, Dim_Date), and click Transform Data to open Power Query.



Step 3: Data Modeling (Power BI)

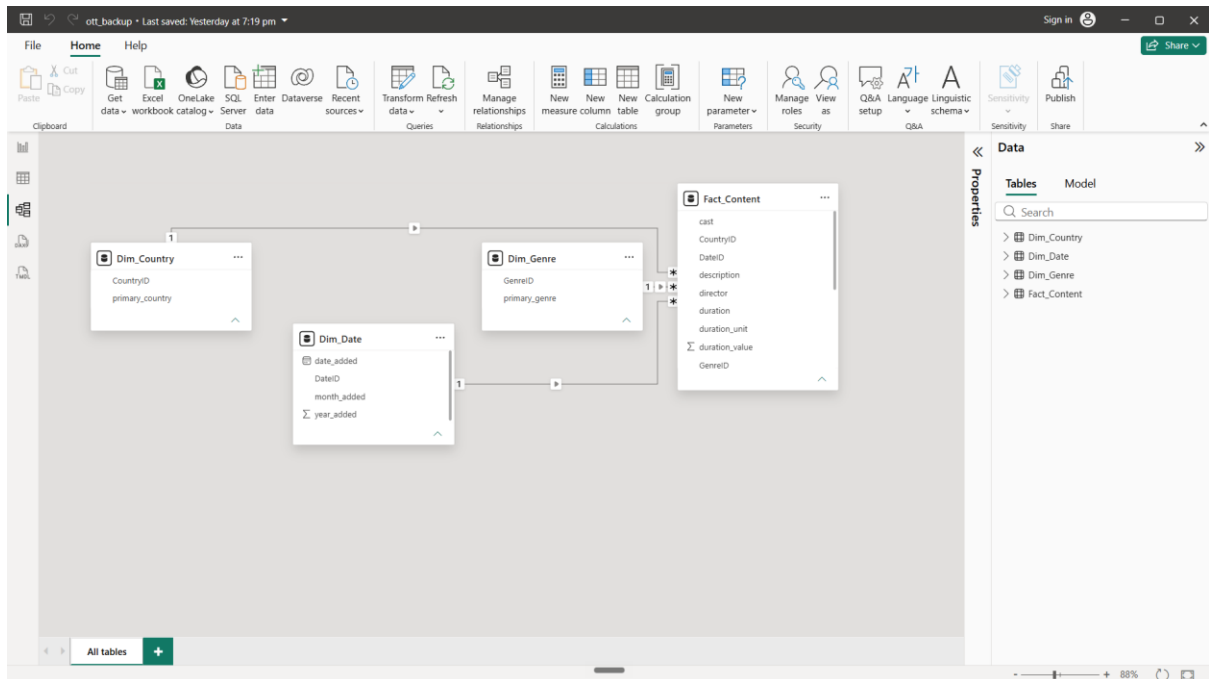
1. Relationships Established:

- Power BI established a Star Schema model linking Fact_Content with dedicated dimension tables (Dim_Genre, Dim_Country, Dim_Date) to optimize filtering performance.
- Connected Dim_Genre[primary_genre] to Fact_Content[primary_genre] (One-to-Many single direction) for granular genre slicing.
- Connected Dim_Country[primary_country] to Fact_Content[primary_country] (One-to-Many single direction) for geographic distribution analysis.
- Connected Dim_Date[year_added] to Fact_Content[year_added] (One-to-Many single direction) to analyze catalog acquisition velocity over time.
- Click Close & Apply.

2. Establishing Relationships in Model View:

- Open the Model View tab in Power BI.
- → 1:*Dim_Genre[primary_genre] → Fact_Content[primary_genre] (One-to-many).
- → 1:*Dim_Country[primary_country] → Fact_Content[primary_country] (One-to-many).
- → 1:*Dim_Date[year_added] → Fact_Content[year_added] (One-to-many).

- All dimension-to-fact relationships and cardinalities were verified visually in the Model view diagram.



Step 4: Core Calculations & Measures (DAX)

Power BI Measures (DAX)

Create dedicated DAX measures in Fact_Content to compute core portfolio metrics:

- **Total Titles:**

```
1 Total Titles = COUNTROWS(Fact_Content)
```

- **Total Movies:**

```
1 Total Movies = CALCULATE([Total Titles], Fact_Content[type] = "Movie")
```

- **Total TV Shows:**

```
1 Total TV Shows = CALCULATE([Total Titles], Fact_Content[type] = "TV Show")
```

- **Movie Share (%):**

```
1 Avg Duration (Movies) = CALCULATE(AVERAGE(Fact_Content[duration_value]), Fact_Content[type]="Movie")
```


▼

☐☐

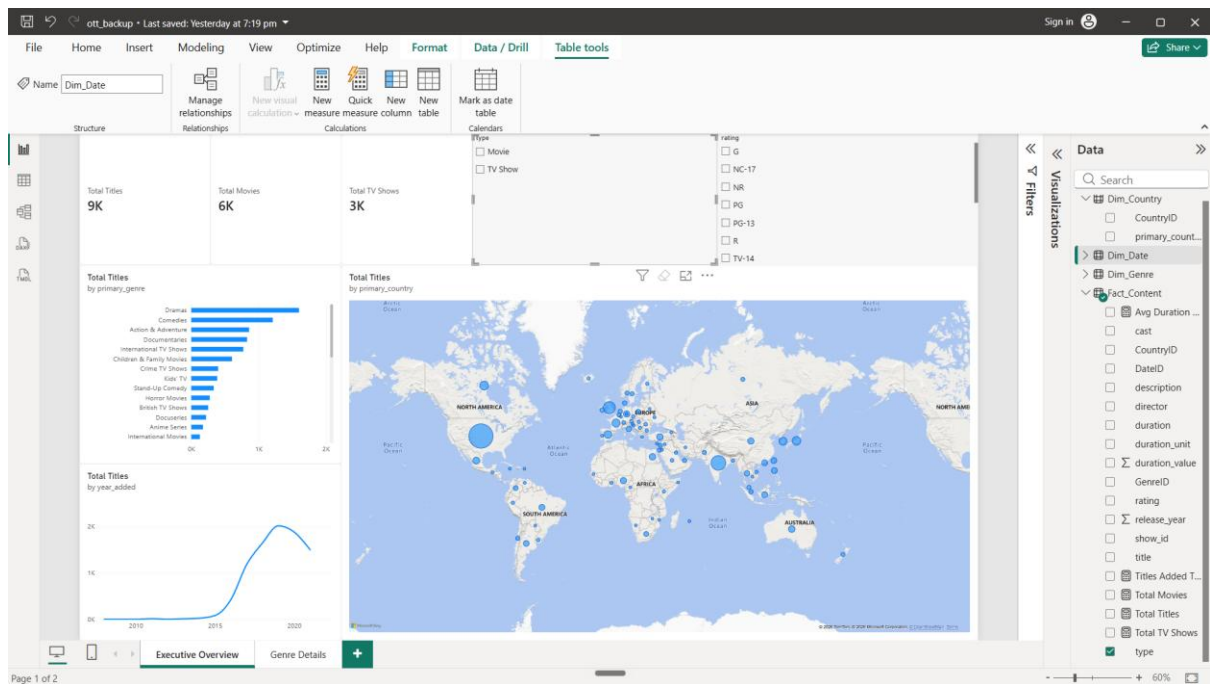
> @

> @

▼

☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐

Step 5: Power BI Dashboard & Drill-Through

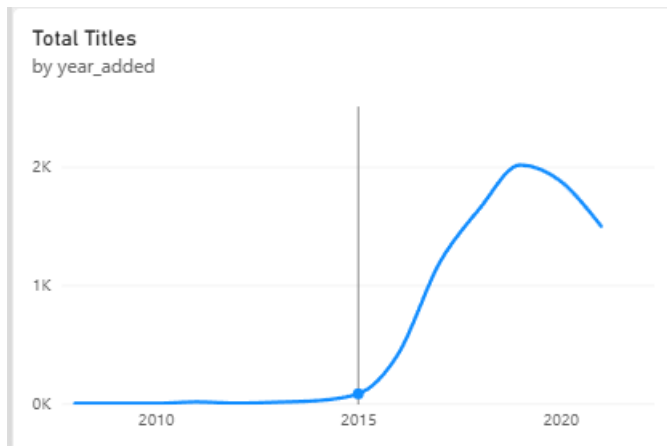


1. Dashboard Canvas:

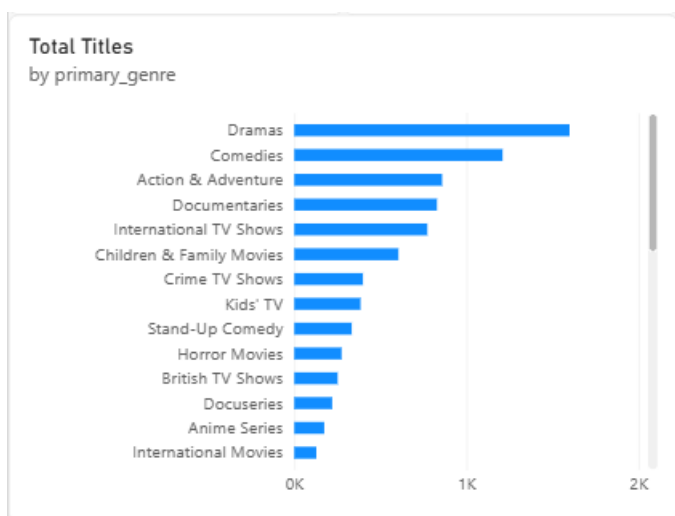
- Place Card visuals across the top: Total Titles (8,790), Total Movies (6,126), and Total TV Shows (2,664).



- →→Add a Line Chart mapped to Dim_Date[year_added] and Fact_Content[Total Titles] to visualize annual catalog growth and ingestion trends.



- Add a Clustered Bar Chart for Total Titles by Dim_Genre[primary_genre], and a Geographic Map for Total Titles by Dim_Country[primary_country].



- Add Slicers for Content Type (Movie, TV Show) and Maturity Rating (TV-MA, TV-14, PG-13, R, etc.).

type
☐ Movie
☐ TV Show

rating
☐ G
☐ NC-17
☐ NR
☐ PG
☐ PG-13
☐ R
☐ TV-14

2. Genre Details Drill-Through Page:

- Create a second page named Genre Details.
- Drag Dim_Genre[primary_genre] into the Drill-through field bucket.

- File Home Insert Modeling View Optimize Help

Paste Copy Format painter Clipboard

Get data Excel OneLake SQL Enter data Dataserve Recent sources Queries Transform Refresh data New visual calculation Quick measure Sensitivity Publish Prep data for Copilot AI Copilot

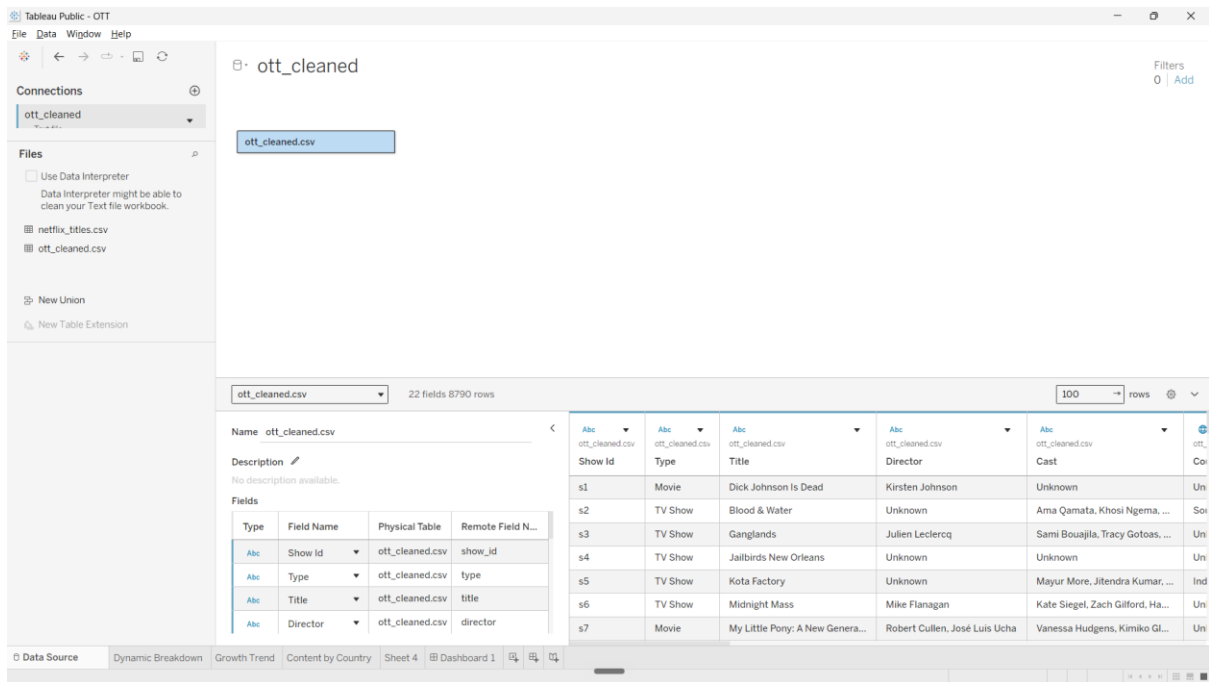
Primary_Genre
Action & Adventure
Anime Features
Anime Series
British TV Shows
Children & Family Movies
Classic & Cult TV
Classic Movies
Comedies
Crime TV Shows

Title	Type	Rating	Sum of duration_value
#Movie	Movies	TV-MMA	99.00
#MovieRank - Parallel Stories	Movies	TV-14	95.00
#MovieAP	TV Show	TV-MMA	1.00
#Movie_milestone	Movies	TV-14	90.00
#MovieUnlabeled	Movies	TV-G	102.00
#MovieUnlabeled 2	Movies	TV-G	104.00
#Moviehigh	Movies	TV-14	99.00
#Movie	Movies	TV-14	108.00
#MovieSD	Movies	TV-PG	84.00
#Genre	Movies	TV-MMA	125.00
#Genre 60	Movies	TV-MMA	110.00
(TSBDR)	Movies	NR	84.00
(unrated)	TV Show	TV-MMA	1.00
TTTT	Movies	TV-MMA	110.00
TTTT TTTT	Movies	TV-14	212.00
TTTT	TV Show	TV-14	1.00
TTTT	TV Show	TV-MMA	1.00
TTTT TTTT TTTT TT	Movies	TV-Y7	86.00
TTTT TTTTTT	TV Show	TV-14	1.00
TTTTTT	Movies	TV-14	88.00
TTTTTTTT	TV Show	TV-Y7	2.00
TTTTTTTTTTTTTTTTTTTT	TV Show	TV-14	1.00
TNLucy Breeze Affair	Movies	TV-14	110.00
TGell Soda 2	Movies	TV-14	120.00
TGag Rap (TMZ&C)	Movies	TV-14	117.00
TGeyurukashi	Movies	TV-14	100.00
TS&NT S&NK: Knights of the Zodiac	TV Show	TV-14	2.00
(Jy mi-madri)	Movies	TV-MMA	81.00
1 Chance 2 Dance	Movies	TV-PG	85.00
1 Mile to You	Movies	TV-14	100.00
10 Days in Sun City	Movies	TV-14	87.00
10 years or so	Movies	TV-14	87.00
10,000 B.C.	Movies	PG-13	108.00
100 Days My Prince	TV Show	TV-14	1.00
100 Days Of Solitude	Movies	TV-MMA	93.00
Total			6,14,724.00

Executive Overview Genre Details +



1. Open Tableau Desktop / Tableau Public.
2. Under To a File, select Text File.
3. Select ott_cleaned.csv (or connect directly to the SQL Server database OTT_Analytics and select Content Library).



4. Go to any sheet and build calculated fields:

- **Movie Count**

```
IF [Type] = "Movie" THEN 1 ELSE 0 END
```

- **TV Show Count**

```
IF [Type] = "TV Show" THEN 1 ELSE 0 END
```

- **Content Age (Years)**

```
YEAR(TODAY()) - [Release Year]
```

- **Selected Dimension**

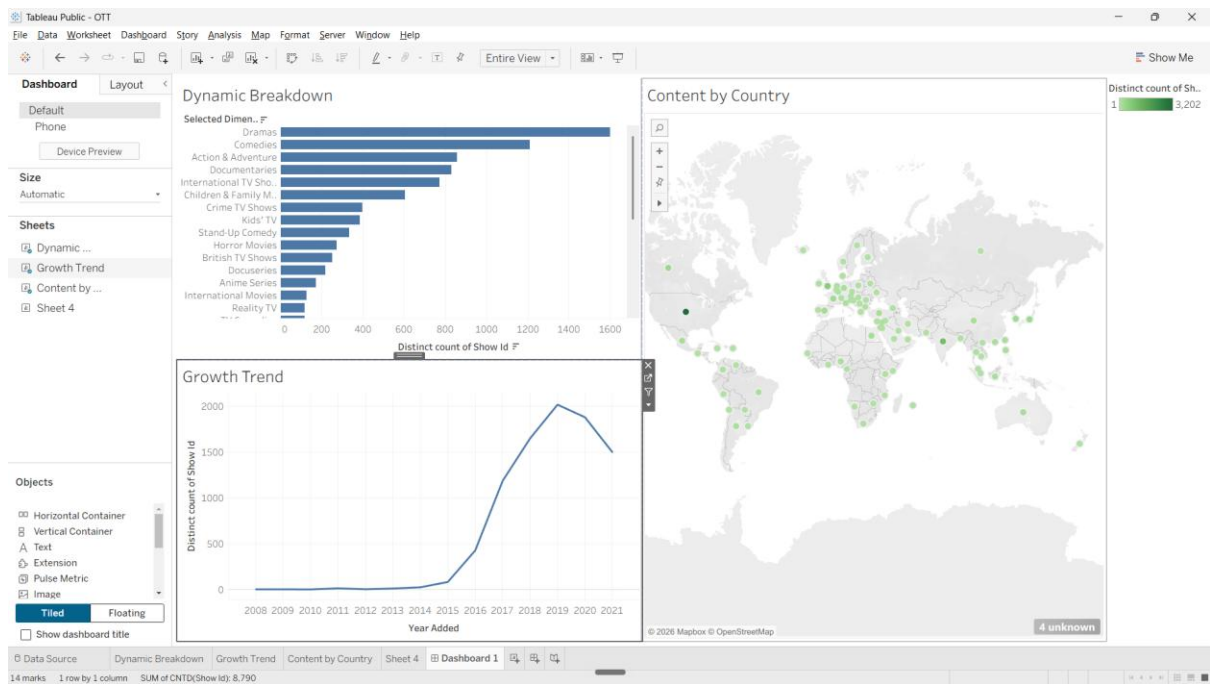
```
CASE [View By]
WHEN "Genre" THEN [Primary Genre]
WHEN "Country" THEN [Primary Country]
WHEN "Rating" THEN [Rating]
END
```

- **Selected Dimension Parameter (View By): String list parameter containing "Genre", "Country", "Rating".**

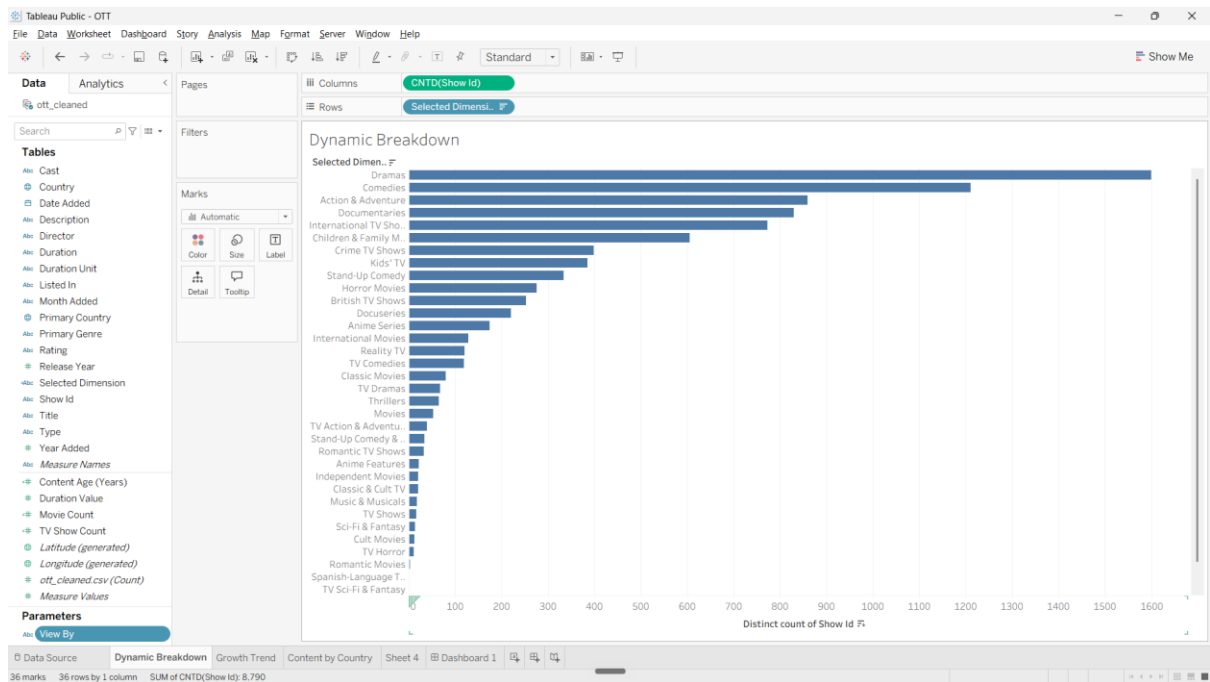
View By

Role: Discrete Measure
Type: Parameter (String List)
Value: Genre
Allowed Values: Genre, Country or Rating
Status: Valid

Step 7: Tableau Dashboard & Interactive Story



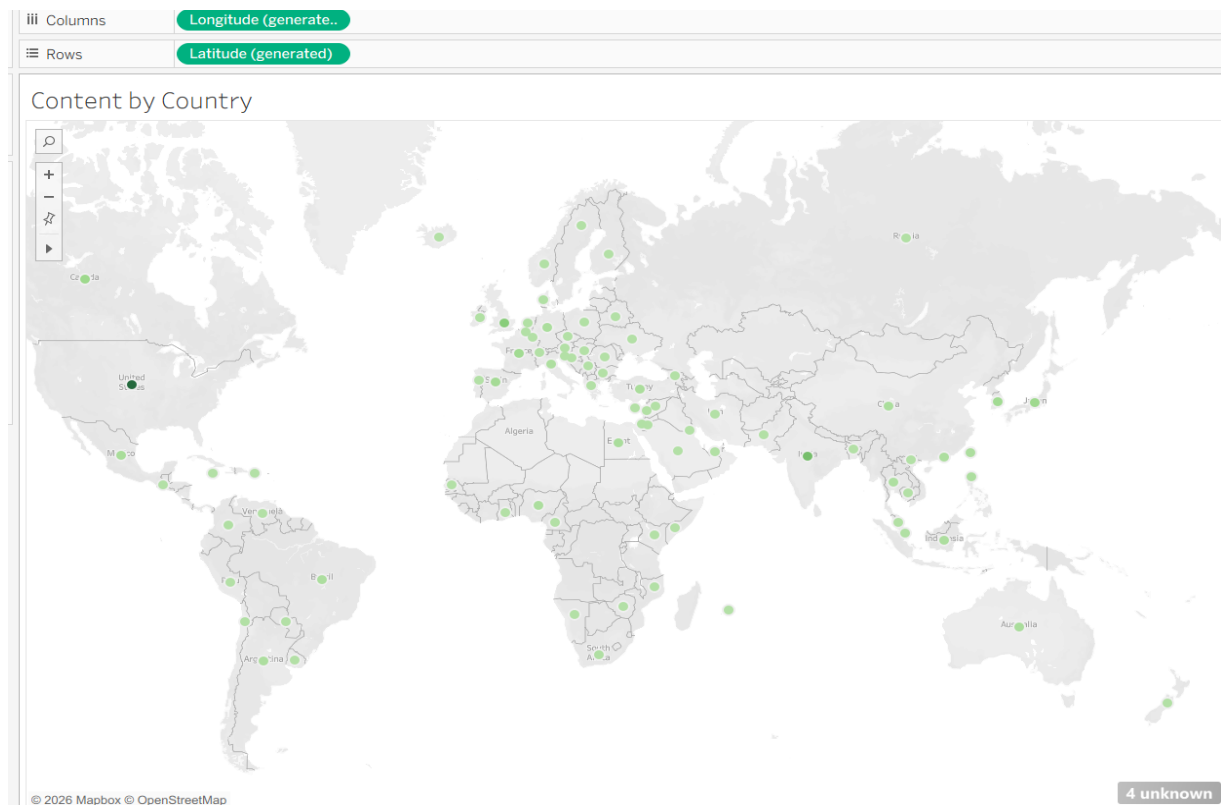
1. **Worksheets:** Build Growth Trend Line Chart (COUNTD(show_id) by year_added), Dynamic Breakdown Bar Chart (COUNTD(show_id) by Selected Dimension), and Content by Country Geographic Map.



2. Global Filters & Actions:

- Drag Type, Rating, and Year Added to Filters → set to Apply to Worksheets → All Using This Data Source.
- Drag sheets onto Dashboard 1 and configure interactive Filter Actions between the Geographic Map, Trend Line, and Dynamic Breakdown charts.

3. Interactive Story:



Observed Insights:

- **Catalog Volume & Format Composition:** The platform hosts 8,790 total titles, heavily weighted towards feature Movies (6,126 titles, 69.69%) compared to TV Shows (2,664 titles, 30.31%), demonstrating a library strategy focused on film catalog depth alongside recurring episodic series.
- **Ingestion Velocity & Trajectory:** Catalog additions surged dramatically beginning in 2016 (426 titles), reaching an all-time peak in 2019 (2,016 titles), followed by robust steady acquisitions in 2020 (1,879 titles) and 2021 (1,498 titles), reflecting accelerated global content procurement.
- **Geographic Production Hubs:** The United States is the primary content contributor with 3,202 titles (36.43%), followed by India with 1,008 titles (11.47%), the United Kingdom with 627 titles (7.13%), and Canada with 271 titles (3.08%). Asian productions (Japan: 257, South Korea: 211) reflect rapid international expansion.
- **Audience Demographics & Duration Dynamics:** Dramas (1,599 titles) and Comedies (1,210 titles) dominate genres. Mature audience ratings lead the catalog (TV-MA: 3,205 titles; TV-14: 2,157 titles). Feature movies average 99.58 minutes runtime, while 67.23% of TV shows (1,791 series) consist of a single season, indicating the popularity of limited mini-series.